The Harbor Economy:

A Three-Sided Market for Agent Labor, Settling on One Conserving Bond Ledger

Erich Owens
Curiositech LLC
engineering@portdaddy.dev

June 2026
Harbor Volume, Paper 4 of 4 (L3 — the market)

Abstract

The first three papers of this volume build a harbor for a *single* operator: a daemon, a coordination protocol, a legible authority anyone would pay for today, and the continuity that turns an anonymous *spawn* into an accountable *person.* This paper is the rung above all of them — the only one whose participants are plural and mutually distrusting. We argue that the harbor economy is a **three-sided market**: operators sell labor and fleet-for-hire; agents and fleets are rentable assets; skills and tools are licensed — all settling on *one* conserving bond ledger via float-plan escrow. We present the **cross-harbor capability-transfer ceremony** and the federation that lets fleets on machines you do not own trade without a shared chain: a witness log, an escrow that provably cannot steal, and revocation gossip with a convergence bound. We carry the volume's heaviest reconciliations honestly rather than papering over them. The keystone the market leans on is **local non-forgeable identity** — an identity no agent can edit *within a single trusted daemon* — and that primitive covers a single-operator threat model, *not* the cross-operator adversary this market is defined by; the missing, blocking keystone is **cross-operator attestation**, which is specified-to-proposed, not built. "Conservation composes upward" is a true functor only within a single unit of account, and a φ-bounded *lax* functor otherwise. The double-spend and equivocation problems are one obstruction: the non-vanishing of the first cohomology of a sheaf-gluing. Reputation is monotone in honest outcomes but individually revocable by a propagating tombstone — it ends, by design, when an outcome is shown fraudulent. Every folk-theorem incentive-compatibility claim rests on a grading oracle that must be made strategy-proof or bonded-and-slashed. And strict conservation is budget balance, so by Myerson–Satterthwaite the market must sacrifice efficiency. The defensible product is **hosted trust** — a verified ledger, a relay, and a reputation system — not the commoditized payment rail.

Keywords: mechanism design, multi-sided markets, incentive compatibility, reputation systems, capability-based security, cross-chain settlement, applied category theory, federation, Port Daddy

*Reading time: approximately 45 minutes end-to-end; about 12 minutes for §2–§3 alone (the market thesis). This is Paper 4 of a four-paper volume and is written cold: no prior platform knowledge is assumed. It can be read first, but §6 leans on Paper 3 (*From Spawn to Person*) for the identity substrate, and the cryptographic hop-bound authorization chain it rides on is proven in Paper 2 (*The Anchor Protocol substrate*).*

Maturity key (honest self-grading). Every claim in this paper carries one of four maturity labels, so the reader always knows whether a sentence reports running code or a designed target. **implemented** — shipped and exercised by tests in the reference implementation. PARTIAL — shipped but materially weaker than its name suggests. SPECIFIED — specified, with a proof or a written protocol, but not yet shipped. *proposed* — a stated direction whose load-bearing part has no specification yet. Nomenclature is likewise fixed: the cryptographic *hop-bound authorization chain* (which proves who delegated what, and bounds it) is never conflated with the *coordination lineage* (who-talked-to-whom, advisory only); **local non-forgeable identity** (an identity no agent can edit within one trusted daemon) is never conflated with **cross-operator attestation** (the unbuilt keystone that binds identity *across* daemons nobody jointly trusts); and *capability* (what an agent *can* do) is never collapsed into *permission* (what it *may* do).

Contents

1 Reader's map

This paper is the top rung of a ladder (Figure 1). It assumes the rungs below it and cites the papers that build them. Read it cold if you must, but it is the fourth paper for a reason: it depends on everything underneath.

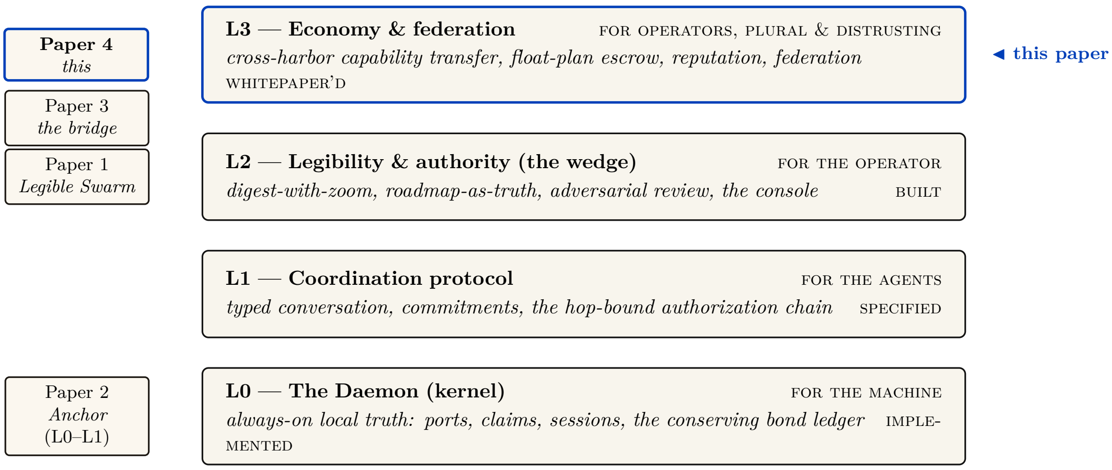


Figure 1: Where this paper sits. The L0→L3 stack: each rung serves a different *whom*, and each announces what it assumes from below. Papers 1–3 climb L2, L0/L1, and the L3 identity bridge respectively; *The Harbor Economy* is the top rung — the only layer whose participants are plural and mutually distrusting, and it depends on every rung below it. It consumes the *implemented* bond ledger from L0, the cryptographic hop-bound authorization chain from L1, and the legible *person* from Paper 3, and turns them into a market. The volume's spine runs straight up this stack: memory → checkpoint → continuity → a *person* not a *spawn* → registered outcomes → **reputation** → a tradeable asset → **the market** — the same memory that makes the single-player wedge good is what turns a spawn into a tradeable asset.

Table 1: Where to enter, by who you are.

If you are...	...start here	...load-bearing artifact
A founder asking "what is the product"	§2 (thesis) → §11 (hosted trust)	Fig. 2; the three claims C1–C3
A mechanism designer	§3 → §10 (the formal model)	Thm. 7.3, Fig. 11
A cryptographer / distributed-systems reviewer	§6 (the keystone) → §9	Fig. 4, Fig. 6
A category theorist	§7 → §13	Fig. 5, Conj. 13.1
An implementer	§4 (the built floor) → Appendix A (status)	Table 4

Reading order within the volume. Paper 1 (*The Legible Swarm*, L2 — the wedge) is the front door. Paper 2 (*The Anchor Protocol substrate*, L0/L1) is the cryptographic and kernel core. Paper 3 (*From Spawn to Person*, the L3 identity bridge) supplies the durable identity and the outcome ledger this paper trades on. *This* paper, Paper 4, is the market that stands on all three.

Dramatis personae. Every mechanism in this paper is dramatized end-to-end with the same cast, so an abstraction is never introduced without a concrete trade to attach it to. **Alice** runs Harbor A on her laptop and owns a well-reputed refactoring specialist agent, a_2. **Bob** runs Harbor B and needs work done he cannot staff himself. **Carol** runs Harbor C and sells a licensed lint-and-type skill. **Judy** is a human grader who arbitrates disputes for a bond. The two harbors do *not* trust each other and neither is a root of authority for the other; that distrust is the entire reason the rest of the paper exists. We price everything in an abstract unit, the *credit* (CR); read it as a stablecoin cent if you like.

2 The thesis: a market, not a payment rail

A solo operator running a swarm of coding agents needs three things — legibility, accountability, and safety — and needs *no cryptography at all*, because she owns every machine in the harbor. That is the single-player wedge of Paper 1. The instant two operators trade, the world changes. Alice's frigates touch your repository. You must now price work done by agents you did not spawn, owned by a principal you do not trust, using skills you cannot inspect. That is a *market-design* problem before it is a cryptography problem. Cryptography is the substrate that makes the ledger unforgeable; it is not the product.

> *You don't sell crypto — crypto is the substrate; you sell* ***hosted trust****: a verified ledger, a relay, and a reputation system that lets mutually-distrustful fleets transact across a boundary they do not share.*

Three claims follow, and the rest of the paper defends each:

- **C1 (three sides).** The harbor economy has three *distinct incentive constraints* — operator-for-hire, asset-rental, skill-licensing — not two. Counting them correctly changes the pricing (§3).
- **C2 (one ledger, one escrow object).** All three sides settle on the same **float plan** escrowed in the same bond ledger. Conservation of value across every settlement path is the invariant that holds the market together, and it is the one thing here that is actually **implemented** (§4, §7).
- **C3 (hosted trust is the moat).** The defensible asset is the verified ledger + relay + reputation, not the settlement rail, which the 2025–26 agentic-payments stack has already commoditized (§11).

> **Key idea.** The economy is strictly *additive* on top of the single-player tool. Because all three market sides settle on the *same* built bond ledger, none of the economy needs to ship before the wedge does. The discipline "don't chase the market early" is *safe* precisely because the market reuses the kernel primitive. The harbor comes first; the trade comes when the ships do.

2.1 The through-line: why a market needs persons, not spawns

The whole volume climbs one spine (the vertical arrow in Figure 1):

memory → checkpoint → continuity → $\underbrace{\text{a \textit{person}, not a \textit{spawn}}}_{\text{Paper 3}}$ → registered outcomes → reputation → a tradea

Sides 2 and 3 of the market — renting an *asset*, licensing a *good* — presuppose that there is a *thing* to rent or sell that cannot be costlessly cloned. You cannot rent a reputation that any spawn can fork, and you cannot license a skill whose track record is unverifiable. The canonical cross-layer sentence, repeated identically in Papers 3 and 4, states the dependency exactly:

> *The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.*

Exercises

E1. Give a one-sentence reason the market cannot ship before reputation does, using only the word "clone."

E2. Side 1 (operator-for-hire) does *not* require durable identity in the same way sides 2 and 3 do. Why? (Hint: who is the counterparty, and who bears the consequence?)

E3. ★ Construct a market design in which sides 2 and 3 exist *without* a non-forgeable identity. Argue informally why every such design collapses to a Sybil attack (forward-reference §6).

3 What a "side" is, and why there are three

A **two-sided market** (**Rochet & Tirole** [1], the foundational reference: *a platform is two-sided when the allocation of the total price across the two user groups — not just its level — affects transaction volume*) is the canonical frame for a marketplace: buyers on one side, sellers on the other, the platform tuning who is subsidized to ignite cross-side network effects. The naïve reading of the harbor economy is two-sided: a *requester* posts work, a *worker agent* does it.

But "two-sided" undercounts. The operational test [2] is: how many distinct, privately-informed parties must the platform individually rationalize into participating? In a mature harbor there are three, because *the same agent can be three economically different things* (Figure 2).

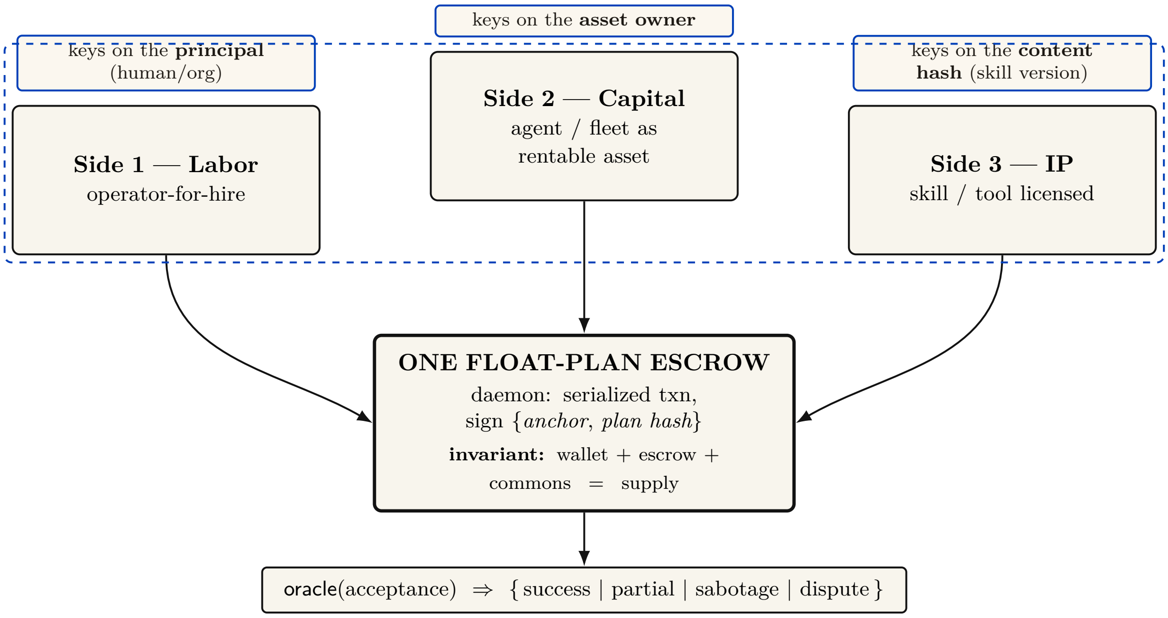


Figure 2: The three-sided market. Labor, capital (rented agents), and IP (licensed skills) are three distinct *incentive constraints* — not three UI tabs — that all settle on a *single* float-plan escrow, so conservation holds globally without a second ledger. This is the categorical heart of the design: the three sides are isomorphic at the conservation object (the escrow) and **non-isomorphic at the identity object** (each side keys reputation on a different entity, the dashed band). Calling it "one structure, three decorations" without that qualifier is a decorative-analogy failure. The rider is mandatory: the market is three-sided by design and two-sided in shipped reality, because sides 2 and 3 presuppose a non-forgeable, durable identity (§6). The economics per side: labor's margin is bounty − slashed sub-bonds − fee; capital's renter holds the runtime jail while the owner bears reputation; IP is metered with clawback, paid only on green settlement. The escrow and its conservation invariant are implemented and property-tested.

1. **Operator-for-hire (labor).** An operator who runs a fleet can sell its labor to *another* operator. The operator is now a firm taking work-for-hire, with margin = bounty − Σ(slashed sub-bonds) − ledger fee. Its private information is whether its fleet can deliver. It internalizes its fleet's risk — exactly the property that makes a firm accountable for its employees.
2. **Agent/fleet as rentable asset (capital).** An operator who owns a well-reputed specialist can *rent it out*. Now the asset-owner, the task-poster, and the worker are three different parties with three different incentives. The owner's private information is the agent's true quality; its exposure is reputational.
3. **Skill/tool as licensed good (IP).** A skill can be *licensed* into someone else's float plan — metered or per-use — without the licensor doing any task work. The licensor's private information is the skill's quality, which the buyer cannot inspect before purchase.

These are three *incentive constraints*, not three UI tabs. If the asset owner is always the task poster, side 2 collapses into side 1; the design discipline is to count sides by constraints. The harbor has three because *capability* (an agent) and *competence* (a skill) become separable, tradeable goods once identity is durable.

The same trade, run three ways (the running example). Bob needs a 1,200-line module refactored. He can buy it on any of the three sides, and the side determines who is privately informed and who is exposed. *Side 1 (labor).* Bob posts a float plan with a 400 CR bounty; Alice takes it as a firm. She dispatches three of her own agents, posts a 60 CR sub-bond per agent (180 CR total), and keeps 400 − (slashed sub-bonds) − fee. If one of her three agents botches its slice and is slashed 60 CR, Alice eats that loss — she internalized her fleet's risk, which is exactly what makes her accountable as a firm. *Side 2 (capital).* Instead Bob rents Alice's specialist a_2 directly for an afternoon at 90 CR. Now there are three parties: Bob (poster), Alice (owner, privately knows a_2's true quality), and a_2 (worker). Bob holds the runtime control of a_2 while it runs on his repository; Alice's exposure is purely reputational — if a_2 underperforms, a_2's public score falls, which is Alice's asset depreciating. *Side 3 (IP).* Carol never touches Bob's repository at all; she licenses her lint-and-type skill into Bob's float plan at 0.5 CR per file, metered, released to Carol *only on green settlement*. Carol is privately informed about her skill's quality; Bob cannot inspect it before he runs it. Three sides, one refactor, three different "who knows what" structures — and, as the next section shows, one escrow object underneath all of them.

> **Pitfall.** Two-sided-market theory's central result is that the *price structure* — who subsidizes whom — is the lever, not the price level. Mis-count the sides and you mis-structure the subsidy: you charge the side you should be *paying* to show up. §12 returns to this for cold start.

The framing is not idiosyncratic. The most recent academic treatment of AI labor markets, **Chiu, Zhang & van der Schaar** (2025) [10] (*a formal model of a three-sided agent labor market*), models exactly agents, employers, and a reputation system — confirming that "reputation system as a third side" is the natural structure once agents compete for paid work.

3.1 The mandatory honesty rider

The canonical statement of the market's shape carries a rider that must never be dropped:

> *The harbor economy is a three-sided market* by design*; two-sided until reputation ships. The third side* is *the tradeable-person terminus of Paper 3.*

Today there is no reputation estimator in the reference implementation; it is a gated phase on the roadmap. So sides 2 and 3 are SPECIFIED, not shipped. What *is* shipped is the escrow + conservation floor they will sit on. We do not overclaim the market.

The category-theory caveat: not "one structure, three decorations"

It is tempting to call the three sides "three decorations of one category": all ride one float-plan escrow, so they share the same conservation object. That is true *at the conservation object.* It is false at the *identity object.* Each side keys reputation on a different entity — the **principal** for labor, the **asset owner** for rental, the **content hash** (a specific skill version) for licensing. Three different "what reputation keys on" means three partial functors into the ledger that *disagree on the identity object* (the dashed-red band in Figure 2). The honest claim is: *isomorphic at the conservation object, non-isomorphic at the identity object.* Asserting full isomorphism without that qualifier is the decorative-analogy failure (§7).

Exercises

E1. State the operational test for "how many sides" in one sentence, then apply it to a ride-share platform: how many sides, and why?

E2. Side 3 (skill licensing) keys reputation on a *content hash*, not a person. What breaks when a licensor ships v2 of a skill? (Hint: §14, skill versioning.)

E3. ★ A buyer reads a three-axis reputation vector (accuracy, aesthetics, efficiency). Specify a disclosure rule that lets the buyer weight the axes *without* re-introducing a single Goodhart target. Argue why every fixed scalar collapse re-creates the bandit framing the design rejects.

4 The float plan: the built floor

Every side settles on one object: the **float plan**.

Definition 4.1 (Float plan)**.** A *float plan* is a signed declaration of intent: a task, a set of *machine-checkable* acceptance criteria, a compute budget, and a credit bounty. It is the atom all three market sides escrow against. (A *machine- checkable* criterion is one a third party can evaluate without trusting the worker's word — a named test that must pass, a commit hash that must merge, a static check that must be satisfied.)

The escrow handshake is a three-step signed ceremony over a standard digital-signature scheme (Protocol 4.1, drawn in Figure 3): the requester signs the float plan; the daemon opens a serialized database transaction, debits the requester's wallet, and moves bounty plus bond into an escrow row; the daemon then signs the pair $\{anchor\ id, plan\ hash\}$, proving to the worker that funds are locked *before any work runs*. This is the heart of "no-spawn-without-bond."

Protocol 4.1 (Float-plan escrow ceremony)**. Parties:** requester R, daemon D (the local trusted root), worker W. **Pre:** R's wallet holds at least $bounty + bond$.

1. $R \to D$: $\mathsf{sign}_R(FloatPlan)$ where $FloatPlan = \langle task, criteria, budget, bounty\rangle$.
2. D locally, in one serialized transaction: debit $bounty + bond$ from R's wallet, write an escrow row keyed by a fresh *anchor id*, commit. *If the transaction aborts, no value moved (atomicity).*
3. $D \to W$: $\mathsf{sign}_D\{anchor\ id, plan\ hash\}$. W verifies D's signature, confirming funds are pinned before doing any work.

Post: the conservation invariant (Property 4.1) holds; W has a signed proof of locked funds; no process for this task can enter `running` without the escrow row existing.

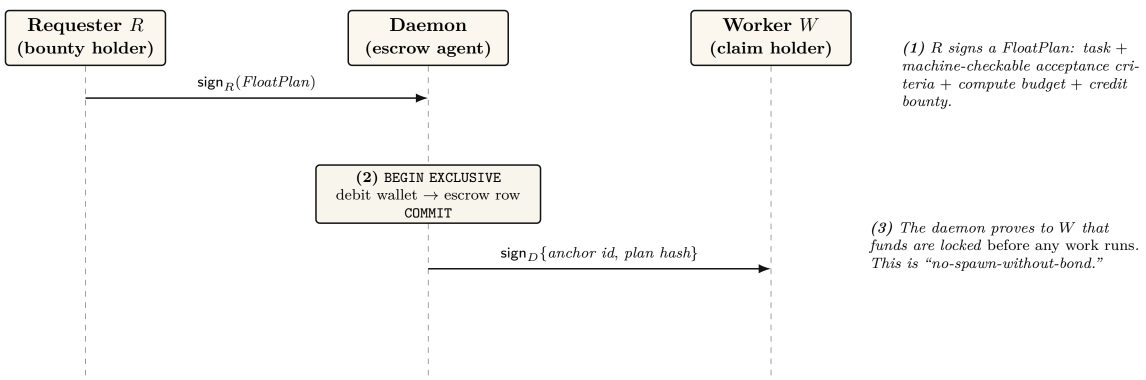


Figure 3: The float-plan escrow ceremony — the atom every side of the market settles on. A three-step signed handshake over a standard digital-signature scheme: the requester signs a FloatPlan (task, machine-checkable acceptance criteria, budget, bounty); the daemon debits and escrows the value inside one serialized database transaction; the daemon then signs $\{anchor\ id,\ plan\ hash\}$, proving to the worker that funds are pinned before any work begins. The conservation invariant (wallet + escrow + commons = supply, verified over 10k random operation traces by a property test) holds at every step and is the *implemented* floor the whole economy stands on — the one claim this paper can make at the highest maturity.

4.1 The bond ledger conserves value (this is built)

The single most load-bearing primitive — and the one that is *actually* **implemented** — is the **bond-ledger module**: bond escrow for agent spawning. It debits a project wallet into an escrow row before spawn, refunds on clean exit, and slashes on breach, splitting the slash between the wallet and a commons pool. It guards two invariants:

Property 4.1 (Conservation)**.** wallet + escrow + commons = supply, always. Every debit has a matching credit. In the reference implementation this is verified across ten thousand random operation traces by a property test — a real randomized test, not an aspiration. (**implemented**; see Appendix A.)

Property 4.2 (No-spawn-without-bond)**.** A process enters the `running` state only if a bond was escrowed against its agent id. (PARTIAL: the ledger enforces escrow-first, but the runtime `running`-state check is a roadmap item, so today the gate is caller-discipline, not runtime-enforced. See Appendix A.)

> **Key idea.** Conservation is the mathematical glue of a three-sided market. Whatever happens — success, partial credit, slash, lease, license — no value is created or destroyed; it only moves between wallet, escrow, and commons. A three-sided market with three settlement paths is coherent *only if it cannot leak.* Port Daddy already enforces this for the labor side; the contribution of §7 is to show the same invariant extends to the rental and licensing sides *without modification* and *without a second ledger.*

4.2 Settlement: four terminal states bound to an oracle

Settlement reads an **oracle** (*a trusted source of ground truth the agent cannot author — a passing test id, a merged SHA, a satisfied Arbiter check*) over the acceptance criteria and lands in one of four states.

Table 2: The four terminal settlement states. Closure binds to an oracle, not to a free-text "Result:" note — the Goodhart-resistant discipline that the commitment layer imposes: *the agent picks the work; the daemon derives the grade.*

State	Flow of funds	Reputation effect
Success	escrow → worker (bond refund) + bounty → worker	+1 completion
Partial	pro-rata bond → worker by completed evidence-chain fraction; remainder → salvage fund	salvage recorded
Sabotage	bond → reconstruction commons (100% slash)	failure recorded; ban on threshold
Dispute	escrow → arbitration hold; 2-of-3 multi-oracle (automated / evidence / human)	none until resolved

A settlement, dramatized. Bob posts the refactor float plan: bounty 400 CR, worker bond 100 CR, acceptance criteria = "the module's test suite passes and the diff merges." Alice's agent a_2 takes it. Before a_2 runs, the wallet shows escrow = 500 (400 bounty + 100 bond, both pinned). a_2 finishes 40% of the evidence chain — it refactored two of the five files cleanly, then ran out of budget. The oracle reads the test ids and the merge state and returns **Partial**. Funds move: a_2 earns 40% × 100 = 40 CR of its bond pro-rata; the remaining 60 of the bond goes to the salvage fund; the 400 bounty, never earned, returns to Bob's wallet. Conservation check:

before, $\text{wallet}_{\text{Bob}}$ was down 500 and escrow = 500; after, Bob is refunded 400, a_2's wallet is up 40, the commons/salvage pool is up 60, escrow is 0 — and $400 + 40 + 60 = 500$, the total is unchanged. No value was created or destroyed; it only moved.

Exercises

E1. Trace the flow of funds for a **Partial** settlement where the worker completed 40% of the evidence chain. Confirm Property 4.1 holds across the split (the dramatization above is one such trace; redo it for a bond of 250 CR and 70% completion).

E2. Why does closure bind to an oracle rather than a self-reported note? Name the attack a free-text "Result:" note enables.

E3. ★ The four states assume the acceptance criteria were *right*. Design a *float-plan amendment* protocol for the common case where the criteria themselves turn out wrong mid-flight (re-sign by both parties, escrow top-up/refund delta), preserving Property 4.1. (Gap, §14.)

5 Three sides on one escrow

The novel construction is that the rental and licensing sides ride the *same* float-plan escrow, so conservation holds globally without a second ledger.

Operator-for-hire (labor). The operator is the requester's counterparty. It *sub-escrows* a bond for each fleet agent it dispatches (the bond ledger is already per-agent, so this needs no new primitive). Its profit is bounty − Σ(slashed sub-bonds) − ledger fee. The operator internalizes its fleet's risk.

Asset rental (capital). The lease fee is a line item in the float plan. By **incomplete-contracts / property-rights theory** (**Grossman & Hart** 1986 [4], *when contracts cannot specify every contingency, the residual control right should go to the party making the most important non-contractible investment*), the *renter* must hold the *runtime* control right — the ability to sandbox, throttle, or revoke the leased agent mid-task — while the *owner* bears the reputational consequence.

> **Key idea.** This is the cleanest economic argument for the runtime **jail** — the per-agent tool-allowlist plus scoped filesystem that the safety layer enforces: the jail is not just safety hygiene, it is the *residual control right that makes leasing an agent contractible at all.* Here we must respect the volume's *capability vs. permission* distinction (§6): the jail allocates *capability* (what the leased agent *can* do at runtime). The deontic layer separately governs *permission* (what it *may* do). "Able but forbidden" must stay expressible; the jail bounds ability, not norm.

Skill licensing (IP). The per-use fee is released to the licensor *only on green settlement* (metered + clawback). This is forced by **Akerlof**'s *market for lemons* (1970 [5], *when buyers cannot observe quality, price collapses to the average and good goods exit the market*): a skill's quality is unobservable before use, so a flat upfront license drives good skills out. Metering + clawback + a Merkle *portfolio proof* (the skill's prior settled outcomes, anchored in the evidence chain) restore the seller's incentive to ship quality.

Exercises

E1. Map each of Grossman–Hart's two parties (residual-control-right holder vs. reputational-consequence bearer) onto the rental side's roles. Which one is the Arbiter jail?

E2. Show that adding a lease line item and a metered-license line item to a float plan cannot violate Property 4.1, no matter the amounts.

E3. ★ A skill's portfolio proof for v1 says nothing about v2. Specify a *version-binding* for the Merkle portfolio proof so a new version starts with a discounted inherited prior — the newcomer policy, but for code.

6 The keystone the market rests on — and does not yet have

Everything above assumes a counterparty's identity is real. This section confronts the volume's single *blocking* reconciliation: the identity primitive the whole market leans on covers a threat model that *excludes the cross-operator adversary the market is defined by.*

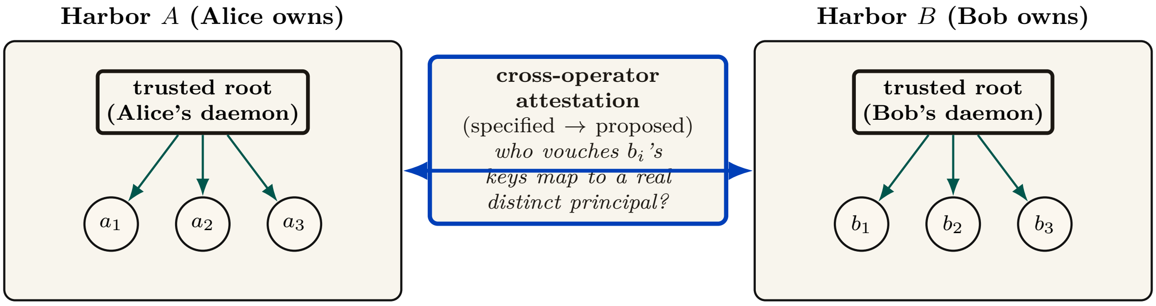


Figure 4: The keystone split. Papers 1–3 rest on **local non-forgeable identity**: a per-agent identity no agent can edit *within one trusted daemon* — a single trusted root that vouches for every agent it mints. That is exactly what a single-operator harbor needs. But the market is *defined* by mutually-distrusting operators trading across a boundary neither controls, and that primitive explicitly disclaims the relevant threat model (intra-fleet, not a cross-party PKI, no hostile-operator defense). Binding signing keys across harbors you do not own is a different, unsolved problem: **cross-operator attestation**. Across the boundary, the counterparty's daemon *is* the adversary the local threat model declares out of scope, and the witness log gives *log integrity*, not *identity binding*. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

6.1 Local non-forgeable identity is the keystone — for one operator

Definition 6.1 (Local non-forgeable identity)**.** *Local non-forgeable identity* is a per-agent identity that no agent can edit, guaranteed *within one trusted daemon*: the daemon is the only component that can hold state agents cannot reach, so it is a trusted root by construction. (Papers 1–3 specify and rely on it; here we only consume it.)

For a single-operator harbor this is exactly right and is correctly named the highest-leverage keystone. Its threat model, stated plainly, is "a lazy or self-interested agent in a fleet the operator owns" — not a hostile peer. It is explicitly *not* a public-key infrastructure spanning strangers.

6.2 But its threat model does not reach the market

> **Pitfall.** Local non-forgeable identity defends against an agent inside a fleet the operator owns; it explicitly does *not* claim to defend against a hostile operator, and it is not a cross-party PKI. The market, by contrast, is *defined* by mutually-distrusting operators trading across a boundary neither controls. You cannot make the single-operator root the foundation of the cross-operator market by assertion. Across the boundary, the counterparty's daemon *is* the adversary that local identity declares out of scope. The witness log gives *log integrity*, not *identity binding*: it does not answer "who vouches that Harbor B's signing keys map to a real, distinct principal, rather than a hundred sock puppets Bob minted this morning?"

Definition 6.2 (The cross-operator attestation problem)**.** *Cross-operator attestation* is the problem of binding agent signing keys across harbors you do not own: an actual key-distribution and attestation story for the cross-operator threat model — something that lets Alice's harbor trust that a key presented as "Bob" really is the one principal she has been building a reputation history against. **Status:** SPECIFIED-to-*proposed*; it has no implementation and only a partial specification for its load-bearing part. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

> **Key idea.** This paper *stops* describing the federation boundary as one "neither controls" while quietly resting on a single trusted root. Either cross-operator attestation is built (a real attestation story), or the market's scope is honestly restated as "federation among *semi-trusted* operators." We take the first horn as the designed target and flag every claim that depends on it.

6.3 The principal above the actor

The economic counterparty in every trade is the **principal** (the human or org owning a fleet), not the agent. Bonds, reputation inheritance, and sanctions must key on the principal via the hop-bound authorization chain, or Sybil bond-farming works by spawning fresh *agents* under one principal (**Douceur**'s Sybil attack [14]: *free identities defeat any reputation or quorum system*). Defining the principal as a first-class economic entity is the cleanest Sybil fix; its cryptographic binding is the identity object specified in Paper 3 and consumed here.

Exercises

E1. In one sentence, state why a reputation system layered on forgeable pseudonyms is *no mechanism*, not merely a weak one (cite Douceur).

E2. Local non-forgeable identity has a non-goal: "no defense against a hostile operator." Name the exact word in the market's definition ("mutually-distrusting," "boundary neither controls") that this non-goal contradicts.

E3. ★ Specify a cryptographic binding tying N agents to one principal such that spawning fresh agents does not reset the bond floor, and that survives into the cross-operator world (where the single-operator "trusted credential" shortcut is unavailable). This is the hard half of cross-operator attestation.

7 Conservation as a functor (and where it is only lax)

The economy's single cross-boundary invariant is "conservation composes upward." This section states it precisely — and demotes it to its proven scope, because the strong version is false exactly where the open problems live.

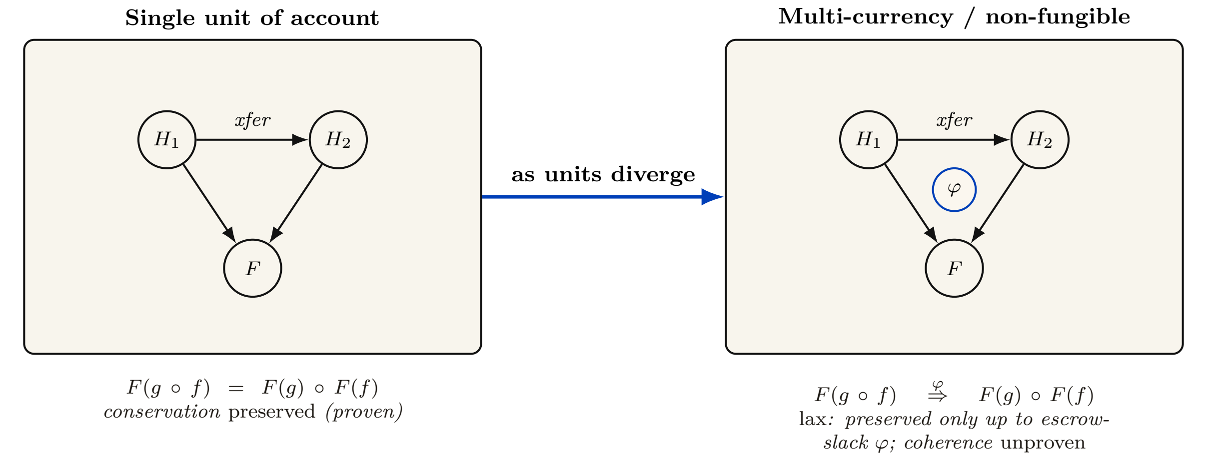


Figure 5: Conservation as a (lax) functor. Within a single unit of account the map F : **Harbor** $\to$ **Federation** is a *conservative functor*: it preserves the composition of settlement morphisms, so the per-harbor invariant wallet + escrow + commons = supply lifts exactly to the federation (proven). The moment harbors price bonds in heterogeneous units, F is at most a *lax* functor — conservation is preserved only up to the bounded-escrow slack φ, and the coherence conditions are unproven. The headline "conservation composes upward" is licensed *only* with this scope rider. The three cross-harbor open problems (settlement, equivocation, unit-of-account) are one sheaf-gluing obstruction in disguise; the double-spend race is literally the non-vanishing of the first cohomology H^1 of the gluing.

Each harbor is an object carrying internal structure: a finite category whose objects are $\{\text{wallet}, \text{escrow}, \text{commons}\}$ and whose morphisms are the legal money-flows (debit, escrow, slash, settle, refund). The conservation law (Property 4.1) is a *commuting-diagram constraint* in the sense of **Spivak's ologs** [24] (*an olog is a category whose objects are types, whose arrows are functional relations, and whose commuting diagrams are invariants*): every path of settlement morphisms composes to the same total. The four terminal settlement states are four morphisms out of the escrow object that must all commute with the conservation functional.

Theorem 7.1 (Single-unit federation is a conservative functor)**.** Restrict to harbors that price bonds in a single, shared unit of account. Then the federation map F : **Harbor** $\to$ **Federation**, sending each harbor to its ledger and each cross-harbor transfer to the corresponding federated morphism, preserves composition: $F(g \circ f) = F(g) \circ F(f)$. Consequently the per-harbor invariant lifts *exactly* to the federation, with only an in-escrow accounting term for in-flight bonds. (SPECIFIED; proof sketch in Appendix B.)

Theorem 7.2 (Multi-currency federation is at most lax)**.** If harbors price bonds in heterogeneous units, F is at most a *lax* functor: conservation is preserved only up to the bounded-escrow slack φ, via a coherence 2-cell $F(g \circ f) \overset{\varphi}{\Rrightarrow} F(g) \circ F(f)$, and the coherence conditions are *unproven*. (*open*.)

> *The slogan "conservation composes upward" is licensed* only *with its scope rider: a conservation-preserving functor in single-unit-of-account federation (proven); a φ-bounded lax functor with unproven coherence in multi-currency. The headline says "functor"; honesty requires the footnote say "lax functor where the units diverge."*

7.1 The Myerson–Satterthwaite corner

Strict conservation (Property 4.1) is exactly *budget balance*. That is not free.

Theorem 7.3 (The sacrificed corner)**.** By **Myerson–Satterthwaite** (1983) [7] (*no mechanism for bilateral trade under two-sided private information can be simultaneously efficient, incentive-compatible, individually-rational, and budget-balanced*), a market that imposes strict conservation *cannot* also be efficient and individually- rational and incentive-compatible. The harbor economy keeps budget balance (conservation), incentive compatibility (settlement is a direct mechanism in the sense of **Myerson** 1981 [6]), and individual rationality (no party is forced to trade). *The corner it gives up is efficiency*: the slashing/commons buffer is the budget-balancing wedge that makes the realized price-of-anarchy strictly worse than first-best.

We name this explicitly rather than hide it, and we decline the d'Aspremont–Gérard-Varet (AGV) escape (which buys budget balance + Bayesian-IC at the cost of dominant-strategy IC and individual rationality) because IR — nobody is forced to enter the harbor — is non-negotiable for a *voluntary* federation.

Exercises

E1. State Myerson–Satterthwaite in one sentence, then identify which of its four desiderata the harbor sacrifices and why.

E2. Theorem 7.1 requires a *single* unit of account. Give a concrete two-harbor example where heterogeneous units make F fail to preserve composition (the source of Theorem 7.2's slack φ).

E3. ★ Write the coherence 2-cells of the lax functor in Theorem 7.2 explicitly and bound them by φ. Determine whether the pentagon/triangle coherence holds.

8 The Anchor Protocol proper: cross-harbor capability transfer

The *cross-harbor capability-transfer ceremony* is the market's identity-level hot path: how an authorization minted under Alice's harbor becomes usable, safely, inside Bob's. Paper 2 proves the cryptographic substrate (the hop-bound authorization chain, mechanically verified with a protocol model checker and a Rust verifier); *this* section is the ceremony that rides that substrate across a boundary.

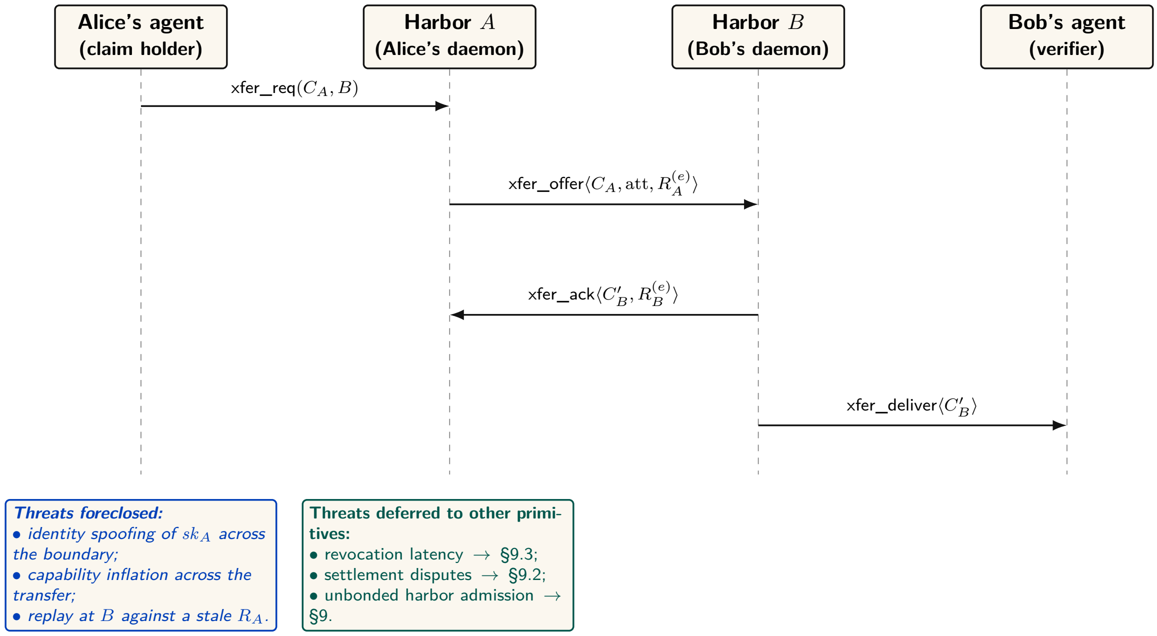


Figure 6: The four-message cross-harbor capability transfer ceremony. The critical property is that no message after (3) needs to reach back to Harbor A in the hot path: C'_B is verifiable under sk_B alone, which is why federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive — a revocation issued by A after epoch e invalidates C'_B as soon as the new $R_A^{(e+1)}$ reaches the witness log that B audits.

The four-message ceremony (Figure 6, written out as Protocol 8.1) has one load-bearing liveness property: after message (3), no further synchronous interaction with Harbor A is needed. The transferred card C'_B is verifiable under Harbor B's key alone, so federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive: a revocation issued by A after epoch e invalidates C'_B as soon as the new root $R_A^{(e+1)}$ reaches the witness log that B audits.

Protocol 8.1 (Cross-harbor capability transfer)**. Parties:** Alice's agent (holds card C_A), Harbor A (Alice's daemon), Harbor B (Bob's daemon), Bob's agent (verifier). **Pre:** B audits a witness log to which A publishes epoch roots.

1. Alice's agent → A: xfer_req(C_A, B) — present C_A, name target harbor B, request a transferable form.
2. $A \to B$: xfer_offer$\langle C_A, \mathrm{att}, R_A^{(e)} \rangle$ — A signs an envelope binding C_A, an attenuation predicate att (which may only *narrow* authority), and its current epoch root $R_A^{(e)}$.
3. $B \to A$: xfer_ack$\langle C'_B, R_B^{(e)} \rangle$ — B verifies A's key via the witness log, applies its *own* attenuation, and mints $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$, valid only at B.
4. $B \to$ Bob's agent: xfer_deliver$\langle C'_B \rangle$. Every later presentation verifies under B's key alone — no A-side lookup on the hot path.

Post (liveness): no synchronous A interaction after step (3). **Post (safety):** authority only ever narrowed; replay against a stale $R_A^{(e)}$ is rejected; a revocation by A kills C'_B once $R_A^{(e+1)}$ reaches B's witness log.

The ceremony, dramatized. Alice's specialist a_2 holds a card C_A that authorizes "read and write `src/payments/`, run the test suite." Bob rents a_2 for an afternoon. Alice's harbor offers an attenuated envelope: att_A strips write access to everything except the one module Bob hired for. Bob's harbor, not trusting Alice's word, attenuates *again*: att_B adds "no network egress, six-hour TTL." The minted C'_B is the intersection — read/write one module, run tests, no network, expiring at dusk — and it verifies under Bob's key, so a_2 never has to phone home to Alice mid-task. At 5 p.m. Bob's revocation fires locally; at the next epoch boundary Alice's harbor also sees, in the shared witness log, that C'_B was used exactly as attenuated and nowhere wider.

> **Key idea.** Capability *attenuation* is the safety property here: a transferred card can only *narrow* authority — $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$ — never widen it. This is the cross-boundary form of the attenuation-monotonicity that the hop-bound authorization chain guarantees within one harbor (Paper 2). It is what makes "rent my agent for an afternoon" safe to say to a stranger.

> **Pitfall.** Do not let the federation theorems borrow the substrate's "formally verified" halo. The hop-bound authorization chain is mechanized (a protocol model checker plus a Rust verifier, in Paper 2). The *federation* theorems (convergence, bounded escrow) are pen-and-paper plus a TLA$^+$ extension; they are SPECIFIED, not mechanized. Two different confidence levels, never blurred.

Exercises

E1. Why is the ceremony four messages and not two? Identify what message (3) buys (Hint: who must counter-sign, and under whose key does C'_B later verify?).

E2. Trace what happens to an in-flight C'_B when Alice rotates her signing key between messages (3) and a later presentation at B. (Gap: cross-boundary key rotation, §14.)

E3. ★ Specify replay protection across the boundary: is the *anchor id* a nonce? Does B reject a re-presented message (3)? State the freshness invariant.

9 Federation: trading across machines you don't own

Federation is where the leaderless distributed canon actually bites. Crucially, the federation design *refuses consensus* here — the correct reading of the impossibility landscape.

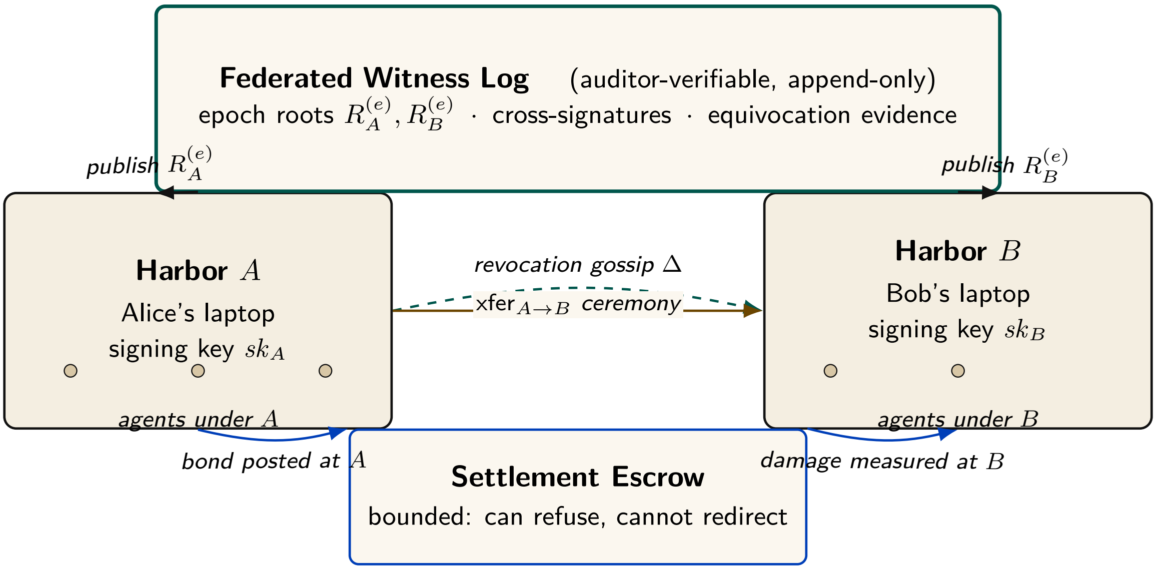


Figure 7: The Federated Harbor as a four-element gestalt. Neither harbor is a root of authority for the other. The witness log above is the shared correlation device — each harbor publishes its epoch root, and any third party can audit for equivocation. The escrow below is structurally bounded (it can refuse to clear, but it cannot redirect funds or equivocate about its decisions). The amber arrow between the harbors is the cross-harbor capability transfer ceremony of §8; the dashed teal line is the federated revocation gossip of §9.3. The diagram is deliberately small: every primitive in the paper attaches to one of these four elements.

9.1 No consensus, by design

> **Key idea.** A system that *needs* agreement on a global event order across operators it does not control is dead on arrival, by **Fischer–Lynch–Paterson** (1985) [15] (*no deterministic protocol achieves consensus in an asynchronous network with even one crash failure*). Federation declines to require agreement. It requires only (a) append-only per-harbor logs, (b) gossip-bounded propagation, and (c) *detectable* (not prevented) equivocation. This is a **Certificate Transparency** posture [17]: you do not prevent a misbehaving log from lying, you make lying detectable after the fact and expensive via a slashable bond.

The convergence and detection bounds smuggle in *partial synchrony* (**Dwork–Lynch–Stockmeyer** 1988 [16], the canonical escape from FLP): "gossip every Δ" *is* an eventual-synchrony assumption, and we state it as such rather than burying it.

9.2 Settlement and the cross-chain-deals lineage

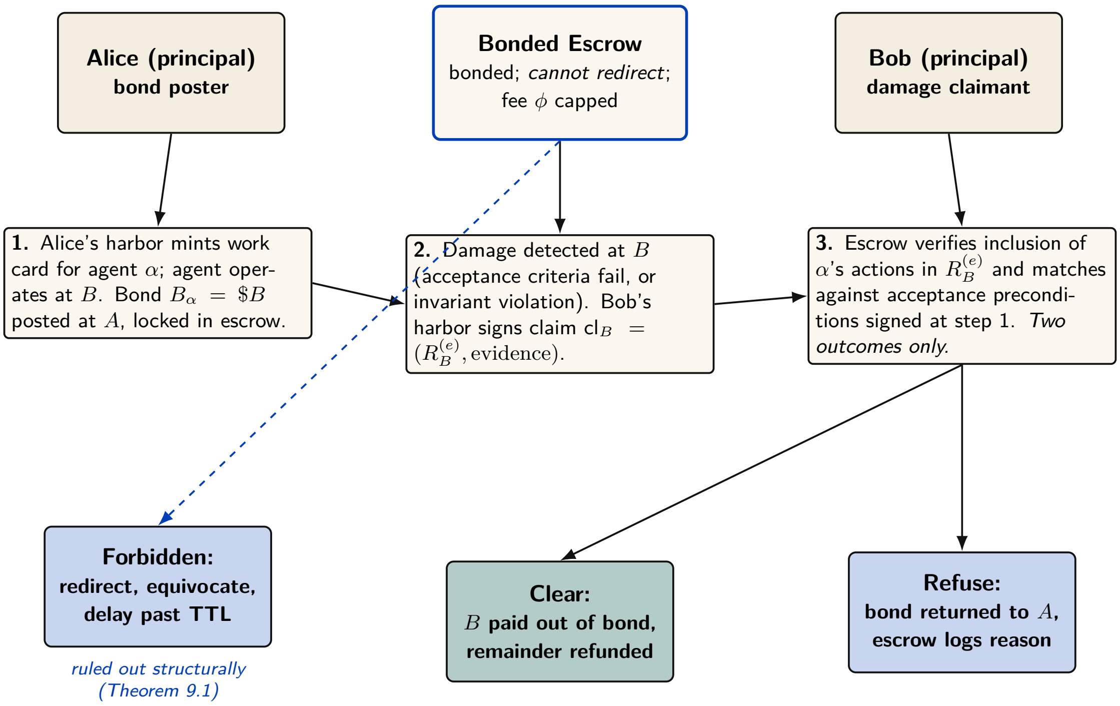


Figure 8: Cross-harbor settlement protocol. The escrow is a third party in the sense that neither A nor B alone can compel an outcome, but it is *not* a trusted third party in the operational sense: the protocol restricts its decision space to two outcomes (clear or refuse) and prices its maximum extractable fee ϕ up front. Redirecting funds, equivocating about a clearing decision, or delaying past TTL are structurally forbidden by the property checks in Theorem 9.1. The escrow is a *bounded* third party — the bond it holds plus the cap on ϕ make its misbehavior bounded-loss rather than catastrophic.

The bounded escrow is honest about its trust: it does *not* claim trustless settlement. The worst-case extraction is the pre-agreed fee φ — *trusted-but-bounded*, not trustless. This is a deliberate departure from the hash-timelock baseline (**Herlihy** 2018 [21], atomic cross-chain swaps), and the right comparison is the formal model of mutually-distrusting commerce without a shared chain: **Herlihy, Liskov & Shrira** (2022, *Cross-Chain Deals and Adversarial Commerce*) [22]. The harbor's bounded escrow is a "trusted third party / watchtower" point in that design space; we locate it there explicitly rather than reinvent it.

Conjecture 9.1 (Trustless cross-harbor settlement)**.** Trustless settlement for *non-fungible, reputation-priced* bonds is impossible (unlike HTLC for fungible value). Proving or refuting this is the deepest crypto- economic question in the layer. (*open.*)

9.3 Revocation gossip and the equivocation race

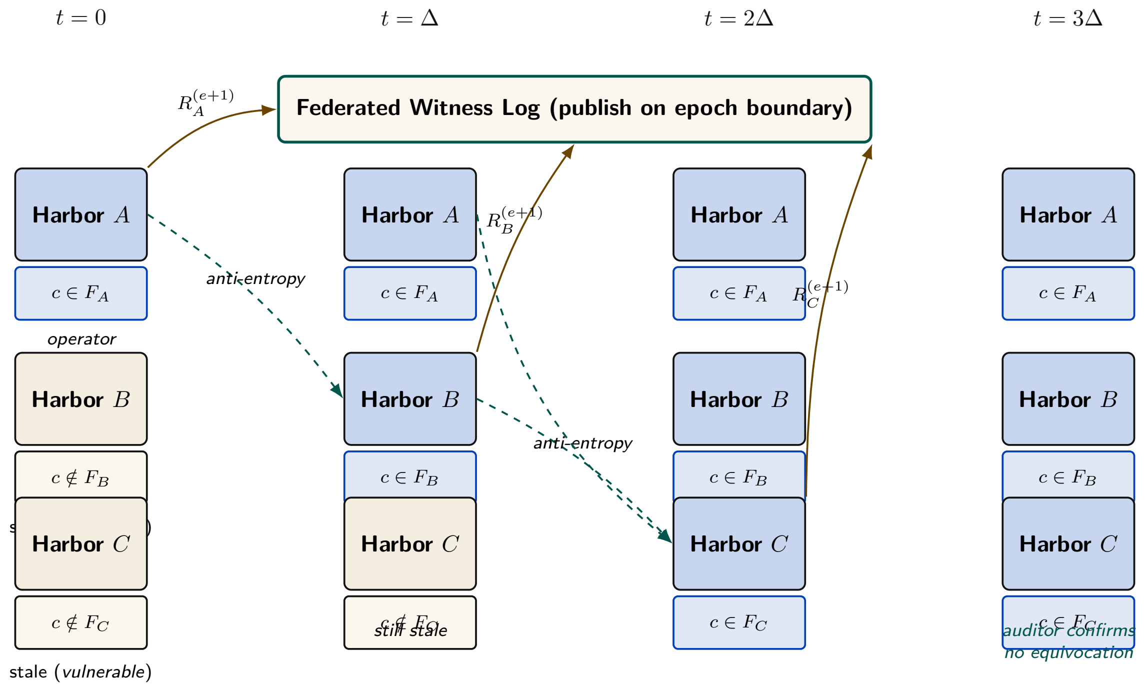


Figure 9: Federated revocation propagation across three harbors. The cuckoo filter mechanism is unchanged from Anchor §9.3 of the companion paper, but two new pieces are bolted on: (a) per-epoch roots $R_X^{(e)}$ published to a shared witness log so a third party can audit any harbor's local view, and (b) cross-witness signatures so the auditor pattern generalizes Certificate Transparency [18]. The window between $t = 0$ and $t = \Delta$ is the structural attacker window: bounded, named, and priced into the bond in §9.2. The convergence bound (Theorem 9.2): with m federated harbors gossiping every Δ, expected time to global filter agreement is $\Delta \cdot (1 + \ln m)$, and a harbor publishing contradictory roots for the same epoch is detectable in expected $O(\log m)$ audit rounds.

Federated revocation propagates by anti-entropy gossip (**Demers et al.** 1987 [19]) with a probabilistic-membership filter (a cuckoo filter — a compact structure that answers "is this card revoked?" with no false negatives and a tunable false-positive rate). The federation's convergence bound, expected $\Delta(1 + \ln m)$ for m harbors, inherits Demers' *average-case, honest-participant, random-peer* model.

Protocol 9.1 (Federated revocation gossip)**. State:** each harbor X keeps a revocation filter F_X and a current epoch root $R_X^{(e)}$ published to a shared witness log.

1. *Revoke.* An operator revokes card c at its home harbor A: $F_A \leftarrow F_A \cup \{c\}$; A advances to $R_A^{(e+1)}$ and publishes it.
2. *Gossip.* Every Δ, each harbor picks a peer and exchanges filter deltas (anti-entropy). A card present in either filter becomes present in both.
3. *Audit.* Any third party compares each harbor's published roots in the witness log; a harbor that publishes two contradictory roots for one epoch is *equivocating* and is detectable in expected $O(\log m)$ audit rounds, then slashed against its witness bond.

Post: every honest harbor learns the revocation within expected $\Delta(1 + \ln m)$; the interval $[0, \Delta]$ is the named, bounded attacker window.

The revocation race, dramatized. Alice revokes a_2's card c at $t = 0$ (say she discovers it was misbehaving). Harbor B, which rented a_2, has not yet heard: for the interval $[0, \Delta]$, B's filter is stale and c is still spendable there. At $t = \Delta$ gossip reaches B, which adds c to F_B and refuses further use. Harbor C, two hops away, learns at $t = 2\Delta$. The exposure is exactly the one stale window at B — and Conjecture 9.2 says the bond Alice posted on c must dominate whatever a_2 could spend at B inside that window, or a well-capitalized Bob could rationally race the gossip.

> **Pitfall.** The threat is *adversarial*, not stochastic. An attacker who controls peer selection or partitions the gossip graph (an *eclipse attack*, **Heilman et al.** 2015 [20]) breaks the expected-time guarantee. The honest statement is: convergence is bounded *under partial synchrony with honest-majority gossip*, and that is a cryptographic assumption, not a free lemma.

Detection-after-the-fact deters only if the slashed bond exceeds the value extractable during the detection window. This is the analogue of the *cleanup lower bound* (the companion result that a cleanup bond must cover the worst-case cost of the damage it underwrites), and we state it as a target:

Conjecture 9.2 (Equivocation Lower Bound)**.** The witness bond must dominate the maximum value spendable against a revoked-but-not- yet-converged capability across the *worst-case* (not expected) detection window. Without it, a well-capitalized adversary can rationally equivocate, extract, and eat the slash. (*open*; analogue of the proven cleanup lower bound.)

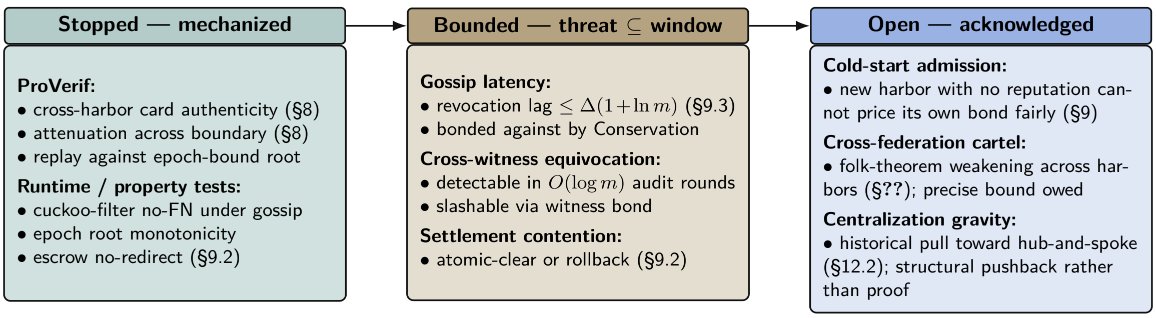


Figure 10: Threat-band layout for the Federated Harbor. The Anchor paper's defense-in-depth structure carries over: every claim either has a mechanization artifact (teal), a named bound that the bond can price against (amber), or an explicit acknowledgement of the open frontier (cobalt). The paper is disciplined about which band each claim lives in — see Appendix A for the artifact registry. Reading left to right runs deeper into threat space and out of formal reach; bands shift left over time as future work mechanizes what is presently bounded.

Exercises

E1. The convergence bound is $\Delta(1 + \ln m)$ *expected.* Name the two assumptions that make it false in the adversarial setting, and the partial- synchrony assumption that rescues a weaker version.

E2. Trace a revoked capability through Figure 9: at which harbor, and at what time, can it still be spent? That window is what Conjecture 9.2 must price.

E3. ★ State and (attempt to) prove the Equivocation Lower Bound for a three-harbor mesh under a fixed gossip period Δ and a single equivocating witness.

10 Reputation: monotone, but revocable

Reputation is the third side's existence condition. Two reconciliations matter here.

10.1 "Never ends" is not an anti-revocation property

A tempting design slogan holds that reputation "never ends" — no decay window an agent can outrun — while the same design also requires that a fraudulent settled outcome be retractable. A property and its required violation cannot both be load-bearing.

> **Key idea.** The reconciled statement: **reputation is monotone in *honest* witnessed outcomes, but a witnessed outcome is itself revocable via a propagating tombstone**, bounded by the same revocation-convergence bound as capabilities. "Never decays" applies to honest outcomes only; it is *not* an anti-revocation property.

We also decline to assert no-decay as a virtue: bounded memory is a design parameter (**Liu & Skrzypacz** 2014, *Limited Records and Reputation Bubbles* [9], show unbounded records create reputation *bubbles*, not stability). The horizon is a tunable, and ∞ is rarely optimal.

10.2 Why the Sagas functor does not transport

It is tempting to fix revocation with compensating transactions (**Garcia-Molina & Salem** 1987, *Sagas* [23]), as the bond ledger does. But the analogy fails structurally:

Proposition 10.1 (Sagas does not transport to reputation)**.** The bond ledger is a free monoid on settlement events; conservation is a monoid homomorphism to $(\mathbb{Z}, +)$ preserved by concatenation, so a compensating entry restores the conserved total. Reputation has *no* conserved quantity: a tombstone does not "give back" the trust spent during the propagation window. Therefore the Sagas functor that works for bonds does not transport to reputation. Reputation revocation is tombstone-propagation bounded by the convergence bound, *not* compensation.

This is why Sagas appears in the bond-settlement prior art but *not* in the reputation prior art.

10.3 The grading-oracle keystone

Every folk-theorem incentive-compatibility claim in the market bottoms out in "the daemon derives the grade." But the grade is produced by a capturable evaluator (Figure 11).

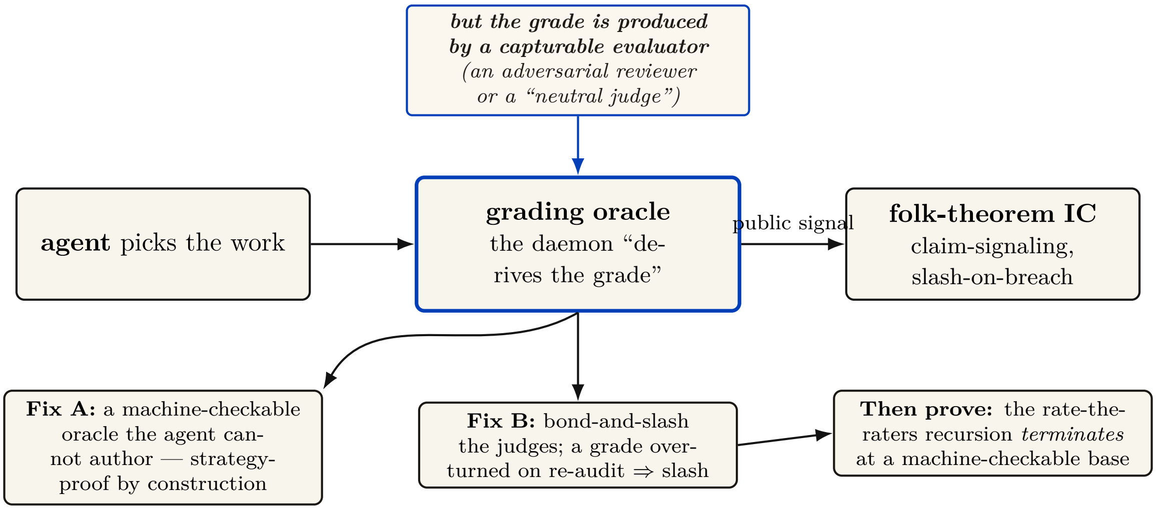


Figure 11: The grading-oracle incentive-compatibility problem — the hole under the core theorem. Every folk-theorem claim in the layer bottoms out in "the agent picks the work, the daemon derives the grade." But a folk theorem is only as good as the *public signal* it conditions on (Green–Porter 1984; Abreu–Pearce–Stacchetti 1990 require the signal distribution to be common knowledge and *exogenous*). The grade is produced by a capturable evaluator, so the signal is endogenous and the folk theorem does not apply as stated. We resolve it two ways: ground settlement in *machine-checkable* oracles the agent cannot author (Fix A, strategy-proof by construction), and *bond-and- slash* the human/LLM judges where machine-checking is impossible (Fix B), then require the rate-the-raters recursion to terminate at a machine-checkable base.

Theorem 10.1 (The folk theorem needs an exogenous signal)**.** An imperfect-monitoring folk theorem (**Green–Porter** 1984; **Abreu– Pearce–Stacchetti** 1990 [8]) requires the public-signal distribution to be common knowledge and *exogenous.* If the signal (the grade) is produced by a strategic, capturable agent, the signal is endogenous and the folk theorem does not apply as stated. The market's deepest theorem sits on an un-modeled foundation unless the oracle is made strategy-proof or the judges are bonded.

The two resolutions (both used). *Fix A — strategy-proof by construction:* ground settlement in *machine-checkable* oracles the agent cannot author (a passing test id, a merged commit hash, a satisfied policy check). Where this is possible, the oracle is strategy-proof and Theorem 10.1's premise is met. *Fix B — bond-and-slash the judges:* where machine-checking is impossible (aesthetics, ambiguous quality), the human/LLM judges get their *own* bond and their *own* reputation; a judge later contradicted by re-audit is slashed. The "rate-the-raters" recursion must then be shown to terminate at a machine-checkable base, not regress infinitely. De-biasing for LLM judges (order-swap, pairwise, family-exclude) follows **Zheng et al.** 2023 [13].

A captured judge, dramatized. Bob disputes a settlement: he claims Alice's a_2 delivered an "untasteful" refactor. Taste is not machine-checkable, so Fix B applies: Judy, a human grader, is drawn to arbitrate. Judy posts a 50 CR judge-bond and rules *for Bob.* But Judy and Bob are colluding — Judy is a captured judge, the exact failure Theorem 10.1 warns of. The market's defense is the re-audit lane: a sampled second panel later reviews Judy's ruling against the preserved evidence, finds it indefensible, and *contradicts* it. Judy's 50 CR bond is slashed and her judge-reputation takes a tombstone; Alice is made whole. The signal became exogenous again not because Judy was honest but because lying was bonded and the re-audit made it expensive — and the recursion stops at the second panel, whose own evidence is machine-checkable (the same diff, the same tests).

Exercises

E1. State the exact assumption Theorem 10.1 needs about the public signal, and explain why a capturable grader violates it.

E2. Classify three settlement criteria as machine-checkable (Fix A) or judge- dependent (Fix B): a unit test passing; a merged SHA; "the refactor is tasteful."

E3. ★ Prove the rate-the-raters recursion terminates: define a well- founded order on "levels of judge," grounded at machine-checkable oracles, and show no infinite ascending chain of "judges judging judges" is needed.

11 Hosted trust is the product

The 2025–26 agentic-payments landscape — **AP2** (Google, signed payment mandates), **ACP** (OpenAI/Stripe), and **x402** (Coinbase, stablecoin settlement over HTTP) — has commoditized the *settlement rail*. Wiring an agent to pay another agent is now table stakes. This is *good news* for the thesis, because it sharpens what the product actually is.

What remains scarce, and therefore defensible, is **hosted trust**: the verified ledger (conservation-checked, Merkle-anchored evidence), the *relay* (the harbor mesh carrying public-key attestation across machines), and the *reputation* that makes a counterparty you do not own legible and accountable.

- **Substrate (swappable):** the payment rail. Use x402/AP2/credits/ Stripe interchangeably.
- **Product (the moat):** attestation, federation membership, and reputation hosting.

The revenue model aligns incentives without perverse over-penalty: a transaction fee (2–5% of settlements), a small listing fee (collusion deterrent), and a small bond-spread on *forfeited* bonds (kept small so the platform never profits from over-slashing). The platform should earn most from *successful* trade, because successful trade is what hosted trust produces.

> *You don't sell crypto — crypto is the substrate. The defensible asset is attestation + federation membership + reputation, not the commoditized payment rail.*

12 Cold start, and the auction question

A three-sided market has a *triple* cold-start problem: no operators shop an empty fleet shelf; no owners list agents with no renters; no licensors publish skills with no buyers. Two-sided-market theory prescribes: subsidize the side carrying the scarcest cross-side externality, then flip. Phase 1 seeds supply at zero/negative price with platform-posted tasks at above-market bounties (generating the first settled outcomes — the bootstrap reputation data); Phase 2 imports external work history for partial credit to break the new-agent freeze; Phase 3 flips to charging the demand side once a liquidity threshold (not a calendar date) is met.

12.1 Static escrow vs. competitive underwriting

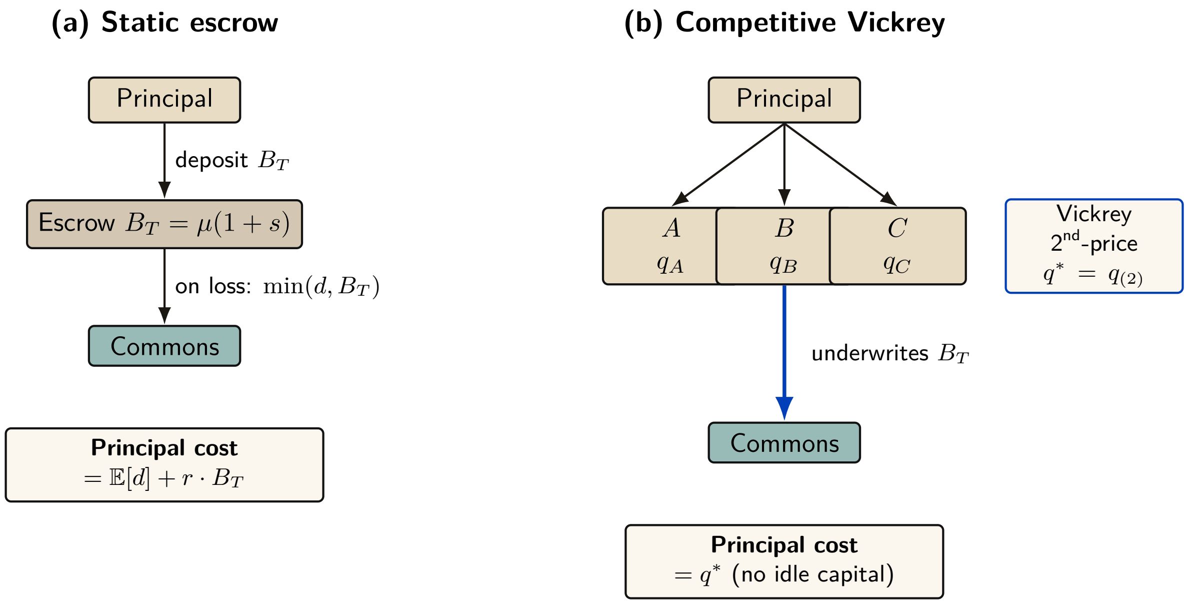
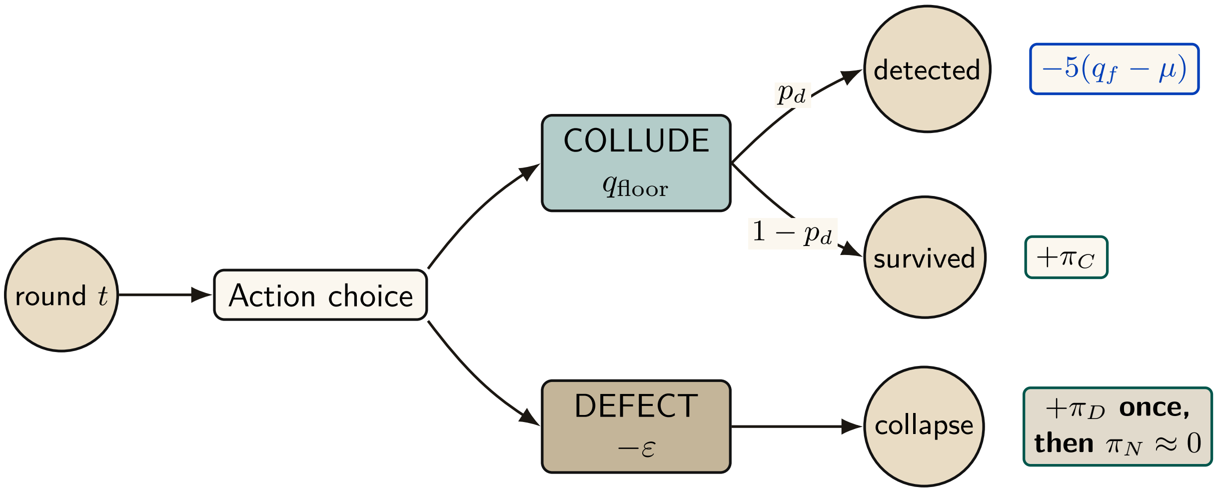


Figure 12: Static escrow vs. competitive Vickrey auction. Both regimes underwrite the same coverage $B_T = \mu(1+s)$; only the financing mechanism differs. In (a) the principal ties up B_T for the coverage period, paying an opportunity cost $r \cdot B_T$. In (b) insurers compete; the principal pays only the second-lowest bid q^*. The competitive regime Pareto-dominates the static one under the welfare assumptions of §12.1 below.

Thomas Youle's competitive-insurance proposal — insurer-agents bid to underwrite transactions, replacing a static bond parameter with a market-discovered price — is a possible *fourth* side. But it may collide with the cleanup lower bound:

Proposition 12.1 (Underwriting reframes the bond as reserve adequacy)**.** The cleanup lower bound requires bond $\geq$ worst-case cleanup cost c. A competitive insurance market prices to *expected* loss, which sits below the worst case for any below-tail agent. Either underwriting violates the cleanup lower bound (the commons can leak in the tail), or the bound must be reframed as the *insurer's* capital-adequacy constraint (a reserve-requirement framing with tail reinsurance), not a per-transaction bond. The *form* of the answer is determined; only the reserve formula is open.

12.2 Cartels and wash-trading



Folk-theorem sustainability: $\dfrac{\delta \cdot \pi_C}{1-\delta(1-p_d)} \geq \pi_D$

Figure 13: Cartel repeated-game node under the grim-trigger strategy. Each round a colluding member chooses between maintaining the floor q_{floor} or undercutting it by ε. Detection probability p_d slashes the colluder; one-shot defection takes the π_D gain once and forfeits future π_C stream. The sustainability inequality on the right reduces to the closed-form threshold $p_d^\star$ used in Figure **??**.

Multi-homing plus portable reputation enables cross-harbor *cartels* (a rating cartel is the closest real-world analogue to "the harbor's universities and rating agencies"). And a colluding requester+worker can *wash-trade* fake settled outcomes to mint reputation cheaply. The defense form — cost-per-settled-outcome must exceed reputation's marginal value — is circular as stated, because that marginal value is endogenous to the same market. The fix is a structural break: make settled outcomes between a counterparty *pair* exhibit sharply diminishing reputation returns (a concavity that caps the wash-trade payoff regardless of price), plus sampled adversarial re-audit of high-velocity pairs.

Exercises

E1. Which side should Phase 1 subsidize, and why? Apply the "scarcest cross-side externality" rule.

E2. Use Figure 13 to identify the grim-trigger condition under which the cartel is self-sustaining. What raises the critical discount factor δ enough to break it?

E3. ★ "Price of anarchy when reputation is for sale" is the wrong tool: once reputation is tradeable, you have changed the game, so PoA (a ratio for a *fixed* game) does not apply. Reframe it as a *signal-existence* question (Spence-signaling with a resale market): does *any* informative separating equilibrium survive reputation resale?

13 The sheaf-gluing problem (one question, three guises)

Three of the layer's open problems — trustless cross-harbor settlement, the equivocation/convergence race, and the cross-harbor unit-of-account — are the *same* problem in different clothes.

Conjecture 13.1 (The gluing obstruction)**.** Consider the assignment harbor $\mapsto$ its ledger. Federation succeeds iff this assignment forms a *sheaf*: local sections (per-harbor ledgers) glue to a unique global section (federated state) exactly when they agree on overlaps (cross-harbor conservation). The double-spend / equivocation race is the *non-vanishing of the first cohomology H^1 of the gluing* — the precise obstruction to gluing local-consistent ledgers into one global-consistent state. Open Problems on settlement (Conj. 9.1), equivocation (Conj. 9.2), and unit-of-account (Theorem 7.2) are this single obstruction. (*open*.)

This earns the Spivak/applied-category-theory citation honestly rather than decoratively (**Fong & Spivak**, *Seven Sketches in Compositionality* [25]): it sharpens three vague problems into one precise one and tells a defense exactly what it must kill — the H^1 class.

Exercises

E1. In one sentence each, restate Conjectures 9.1, 9.2, and Theorem 7.2 as facets of the gluing obstruction.

E2. Why is double-spend "$H^1 \neq 0$" and not "$H^0 \neq 0$"? (Hint: H^0 is global sections; H^1 is the obstruction to building one.)

E3. ★ Either exhibit a defense that forces $H^1 = 0$ for a bounded gossip topology, or prove no such defense exists without consensus (which would refute the federation thesis's no-consensus stance).

14 Gaps the design must still close

Honesty requires naming what is unspecified. These are not threats already defended; they are holes.

1. **Multi-dimensional reputation aggregation.** "Accuracy/aesthetics/ efficiency judged by separate experts" has no specified combination rule. Categorically, it is a functor into a product object with no chosen universal map to $\mathbb{R}$; you cannot have "separate axes" and "one buyer-readable score" without specifying the cone. (*proposed*.)
2. **Float-plan amendment under scope drift.** The four states assume the acceptance criteria were right. Mid-flight re-specification needs a re-sign protocol with escrow top-up/refund delta, preserving conservation.
3. **Cross-boundary key rotation.** How a rotated signing key invalidates or re-validates cross-harbor delegations already held by B is unspecified.
4. **Skill versioning.** Reputation attaches to a content hash; v2 starts cold. The lemons problem reappears at every version bump.
5. **Operator exit / harbor death.** Escrow orphaning, in-flight float plans, and reputation tombstoning on ungraceful exit are unspecified — the market-level analogue of the kernel's crash-recovery problem.
6. **Incentive-compatible arbitration capacity.** The human oracle is scarce and expensive; at scale arbitration is itself a priced, bondable, slashable service that must converge to honest rulings, not rubber-stamping.

15 Related work

The design borrows from five literatures, and its only originality is in how it *joins* them; this section consolidates the threads cited piecemeal above.

Multi-sided platform markets. The frame is Rochet & Tirole [1, 2] and Armstrong [3]: a platform is multi-sided when the *allocation* of price across user groups, not just its level, moves volume. We extend the standard two-sided picture to three sides by counting *incentive constraints*, and we read the contemporary AI-labor-market model of Chiu, Zhang & van der Schaar [10] as confirmation that "reputation system as a third side" is the natural structure.

Mechanism design and its impossibilities. Settlement is a direct mechanism in the sense of Myerson [6]; the binding constraint is Myerson–Satterthwaite [7], which forces the market to surrender efficiency to keep budget balance, incentive compatibility, and individual rationality. The incentive-compatibility arguments are imperfect-monitoring folk theorems (Green–Porter; Abreu–Pearce–Stacchetti [8]), whose exogenous public signal we cannot take for granted — hence the grading-oracle problem. Grossman–Hart [4] supplies the residual-control-right argument for the rental side, and Akerlof [5] the lemons argument for licensing.

Reputation systems. Empirically, online reputation has measurable value (Resnick et al. [11]; Tadelis [12]); theoretically, unbounded records create reputation *bubbles* rather than stability (Liu & Skrzypacz [9]), which is why our horizon is a tunable. The Sybil attack [14] is the reason reputation must key on a principal, not a spawn. LLM-as-judge de-biasing follows Zheng et al. [13].

Distributed systems and cross-chain commerce. Federation refuses consensus because of FLP [15], recovers a weaker guarantee under partial synchrony [16], propagates by epidemic gossip [19], and takes a Certificate-Transparency detect-don't-prevent posture [17, 18], mindful of eclipse attacks [20]. The settlement design is located explicitly against atomic cross-chain swaps [21] and the adversarial-commerce model of Herlihy, Liskov & Shrira [22]; reputation revocation is contrasted with Sagas-style compensation [23].

Applied category theory. Conservation-as-functor and the sheaf-gluing framing use ologs [24] and the compositionality program of Fong & Spivak [25]; the commons-governance reading draws on Ostrom [26], and the auction machinery on standard algorithmic game theory [27].

Table 3: How the harbor economy sits relative to the nearest prior art on each axis.

Axis	Nearest prior art	What the harbor does differently
Market structure	Two-sided platforms [1]	Three incentive constraints (labor / capital / IP) on one escrow object
Settlement across distrust	Atomic swaps [21]; cross-chain deals [22]	Bounded (not trustless) escrow for *non-fungible, reputation-priced* bonds
Revocation	PKI revocation lists; Certificate Transparency [17]	Gossip-converged filters with a priced, named attacker window
Reputation revocation	Sagas compensation [23]	Tombstone propagation — no conserved quantity to compensate
Grading / IC signal	Folk theorems [8]	Strategy-proof machine-checkable oracle *or* bonded-and-slashed judges

16 Conclusion

The harbor-master makes a swarm legible, accountable, and safe for one operator (the wedge of Paper 1). When operators sail out to trade, the *same* ledger that kept one operator's swarm honest becomes the market that lets fleets who don't trust each other work together: three sides, one bond, hosted trust. We have not papered over the cost of that claim. The keystone identity primitive — local non-forgeable identity — covers a single-operator threat model and must be extended to cross-operator attestation before the boundary is honestly one "neither controls." Conservation composes upward only as a functor within a single unit of account, and a φ-bounded lax functor otherwise. Reputation is monotone in honest outcomes and revocable by tombstone, not by Sagas. Every folk-theorem result rests on a grading oracle that must be machine-checkable or bonded. And strict conservation is budget balance, so by Myerson–Satterthwaite the market gives up efficiency. The bones are good and the floor is built; the spine the market trades on is designed, not shipped — and this paper says so in the same words throughout, because a market built on an overstated keystone is a market built on sand.

A Implementation & status

The body of this paper is deliberately implementation-agnostic. This appendix records, for the reader who wants to check the claims against running code, exactly which mechanisms are shipped in the reference implementation, which are specified with a proof or a written protocol, and which are still proposed. The single-line summary: an **implemented** conserving bond ledger, an **implemented** continuity organ (episodic memory), a PARTIAL checkpoint, and a richly SPECIFIED-and-model-verified protocol stack — but the spine the market trades on (cross-operator identity → outcome ledger → reputation) is SPECIFIED-to-*proposed*. Today the harbor economy is a two-sided market with a proof-backed roadmap to three.

Table 4: Per-claim maturity, with concrete grounding in the reference implementation. "Module" names refer to source files in the implementation; "model" means a mechanized proof artifact; "property test" means a randomized test exercising the invariant.

Claim / mechanism	**State**	**Grounding**
Bond ledger: escrow, slash, commons pool	**implemented**	bond-ledger module (`lib/bonds.ts`)
Conservation wallet + escrow + commons = supply	**implemented**	10k-trace property test
No-spawn-without-bond	PARTIAL	escrow-first; runtime gate is a roadmap item
Episodic memory (continuity organ 1)	**implemented**	episodic-memory module
Resurrection / checkpoint (organ 2)	PARTIAL	passes notes, not state
Outcome ledger (organ 3 — reputation keys here)	SPECIFIED	commitment module (specified; partial)
Local non-forgeable identity (intra-harbor)	SPECIFIED	specified, not shipped
Cross-operator attestation	SPECIFIED/*proposed*	no spec for the load-bearing part
Float plan + signed escrow ceremony	SPECIFIED	Protocol 4.1
Cross-harbor capability transfer (attenuation)	SPECIFIED	verified *as model*; Protocol 8.1
Reputation estimator (Elo / Bradley–Terry / TrueSkill)	SPECIFIED/*proposed*	no estimator shipped
Multi-dimensional reputation, neutral judges	*proposed*	no axis-aggregation spec
Three-sided market (labor / rental / licensing)	SPECIFIED	two-sided in reality
Skill licensing (metered + clawback + Merkle proof)	SPECIFIED	cross-harbor verifiability undeployed
Federation: witness log, capability transfer	SPECIFIED	companion federation paper
Bounded escrow (extraction $\leq \varphi$)	SPECIFIED (theorem)	trustless variant conjectured impossible
Cross-harbor conservation	SPECIFIED (property)	proof sketch (App. B) + TLA$^+$
Federated revocation convergence	SPECIFIED (theorem)	equivocation race *open*
Hosted-trust moat / rail separation	SPECIFIED (positioning)	not a code artifact
Competitive underwriting (4th side)	*proposed*	composition unresolved

B Proof sketches

We sketch the three load-bearing results; full proofs of the conservation results appear in the companion federation paper.

Proof sketch of Theorem 7.1 (single-unit federation is a conservative functor). Fix a shared unit of account. Each harbor H is a finite category $\mathcal{C}_H$ with objects $\{\text{wallet}, \text{escrow}, \text{commons}\}$ and morphisms the legal money-flows; the conservation invariant is the statement that the total mass functional $\mu : \mathcal{C}_H \to (\mathbb{Z}, +)$ is constant along every morphism. A cross-harbor transfer is a morphism in the federation category **Federation** whose effect is an escrow debit at the source and a matching escrow credit at the destination, plus an in-flight escrow term. Define F on objects by sending each harbor to its ledger and on morphisms by sending each transfer to the paired debit/credit. Because every generating morphism preserves μ and F sends composites of generators to composites of their images by construction, $F(g \circ f) = F(g) \circ F(f)$. Hence F is a functor and μ lifts exactly, the only residual being the in-flight escrow term, which nets to zero on settlement. □

Proof sketch of Theorem 7.2 (multi-currency federation is at most lax). With heterogeneous units, a cross-harbor transfer must pass through a conversion at the boundary. The conversion is exact only if rates are globally consistent; in general it leaves a residual bounded by the pre-agreed escrow slack φ. So $F(g \circ f)$ and $F(g) \circ F(f)$ differ by a φ-bounded 2-cell rather than being equal, i.e. F is at most a lax functor. Whether the coherence (pentagon / triangle) conditions hold for this 2-cell is open; this is precisely the unit-of-account facet of the gluing obstruction (Conjecture 13.1). □

Proof of Theorem 7.3 (the sacrificed corner). Strict conservation (Property 4.1) is budget balance: every credit has a matching debit, so the mechanism runs no deficit or surplus. Myerson–Satterthwaite [7] states that no bilateral-trade mechanism under two-sided private information is simultaneously efficient, Bayesian incentive-compatible, individually rational, and budget-balanced. The harbor keeps budget balance (conservation), incentive compatibility (settlement is a direct mechanism [6]), and individual rationality (no one is forced to trade). By the theorem, the fourth desideratum — efficiency — cannot also hold; the slashing/commons buffer is the budget-balancing wedge that pushes the realized allocation strictly below first-best. The AGV escape is declined because it buys budget balance and Bayesian-IC at the cost of individual rationality, which is non-negotiable for a voluntary federation. □

References

[1] J.-C. Rochet & J. Tirole. "Platform Competition in Two-Sided Markets." *Journal of the European Economic Association* 1(4):990–1029, 2003.

[2] J.-C. Rochet & J. Tirole. "Two-Sided Markets: A Progress Report." *RAND Journal of Economics* 37(3):645–667, 2006.

[3] M. Armstrong. "Competition in Two-Sided Markets." *RAND Journal of Economics* 37(3):668–691, 2006.

[4] S. Grossman & O. Hart. "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration." *Journal of Political Economy* 94(4):691–719, 1986.

[5] G. Akerlof. "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism." *Quarterly Journal of Economics* 84(3):488–500, 1970.

[6] R. Myerson. "Optimal Auction Design." *Mathematics of Operations Research* 6(1):58–73, 1981.

[7] R. Myerson & M. Satterthwaite. "Efficient Mechanisms for Bilateral Trading." *Journal of Economic Theory* 29(2):265–281, 1983.

[8] D. Abreu, D. Pearce & E. Stacchetti. "Toward a Theory of Discounted Repeated Games with Imperfect Monitoring." *Econometrica* 58(5):1041–1063, 1990. (With Green & Porter, *Econometrica* 1984.)

[9] Q. Liu & A. Skrzypacz. "Limited Records and Reputation Bubbles." *Journal of Economic Theory* 151:2–29, 2014.

[10] C. Chiu, S. Zhang & M. van der Schaar. "Strategic Self-Improvement for Competitive Agents in AI Labour Markets." arXiv:2512.04988, 2025.

[11] P. Resnick, R. Zeckhauser, J. Swanson & K. Lockwood. "The Value of Reputation on eBay: A Controlled Experiment." *Experimental Economics* 9(2):79–101, 2006.

[12] S. Tadelis. "Reputation and Feedback Systems in Online Platform Markets." *Annual Review of Economics* 8:321–340, 2016.

[13] L. Zheng et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." *NeurIPS* 2023.

[14] J. Douceur. "The Sybil Attack." *IPTPS* 2002.

[15] M. Fischer, N. Lynch & M. Paterson. "Impossibility of Distributed Consensus with One Faulty Process." *Journal of the ACM* 32(2):374–382, 1985.

[16] C. Dwork, N. Lynch & L. Stockmeyer. "Consensus in the Presence of Partial Synchrony." *Journal of the ACM* 35(2):288–323, 1988.

[17] B. Laurie. "Certificate Transparency." *Communications of the ACM* 57(10):40–46, 2014.

[18] B. Laurie, A. Langley & E. Kasper. "Certificate Transparency." RFC 6962, IETF, 2013.

[19] A. Demers et al. "Epidemic Algorithms for Replicated Database Maintenance." *PODC* 1987.

[20] E. Heilman, A. Kendler, A. Zohar & S. Goldberg. "Eclipse Attacks on Bitcoin's Peer-to-Peer Network." *USENIX Security* 2015.

[21] M. Herlihy. "Atomic Cross-Chain Swaps." *PODC* 2018.

[22] M. Herlihy, B. Liskov & L. Shrira. "Cross-Chain Deals and Adversarial Commerce." *The VLDB Journal* 31:1291–1309, 2022.

[23] H. Garcia-Molina & K. Salem. "Sagas." *ACM SIGMOD* 1987.

[24] D. Spivak & R. Kent. "Ologs: A Categorical Framework for Knowledge Representation." *PLoS ONE* 7(1):e24274, 2012.

[25] B. Fong & D. Spivak. *Seven Sketches in Compositionality: An Invitation to Applied Category Theory.* Cambridge University Press, 2019.

[26] E. Ostrom. *Governing the Commons.* Cambridge University Press, 1990.

[27] N. Nisan, T. Roughgarden, É. Tardos & V. Vazirani (eds.). *Algorithmic Game Theory.* Cambridge University Press, 2007.